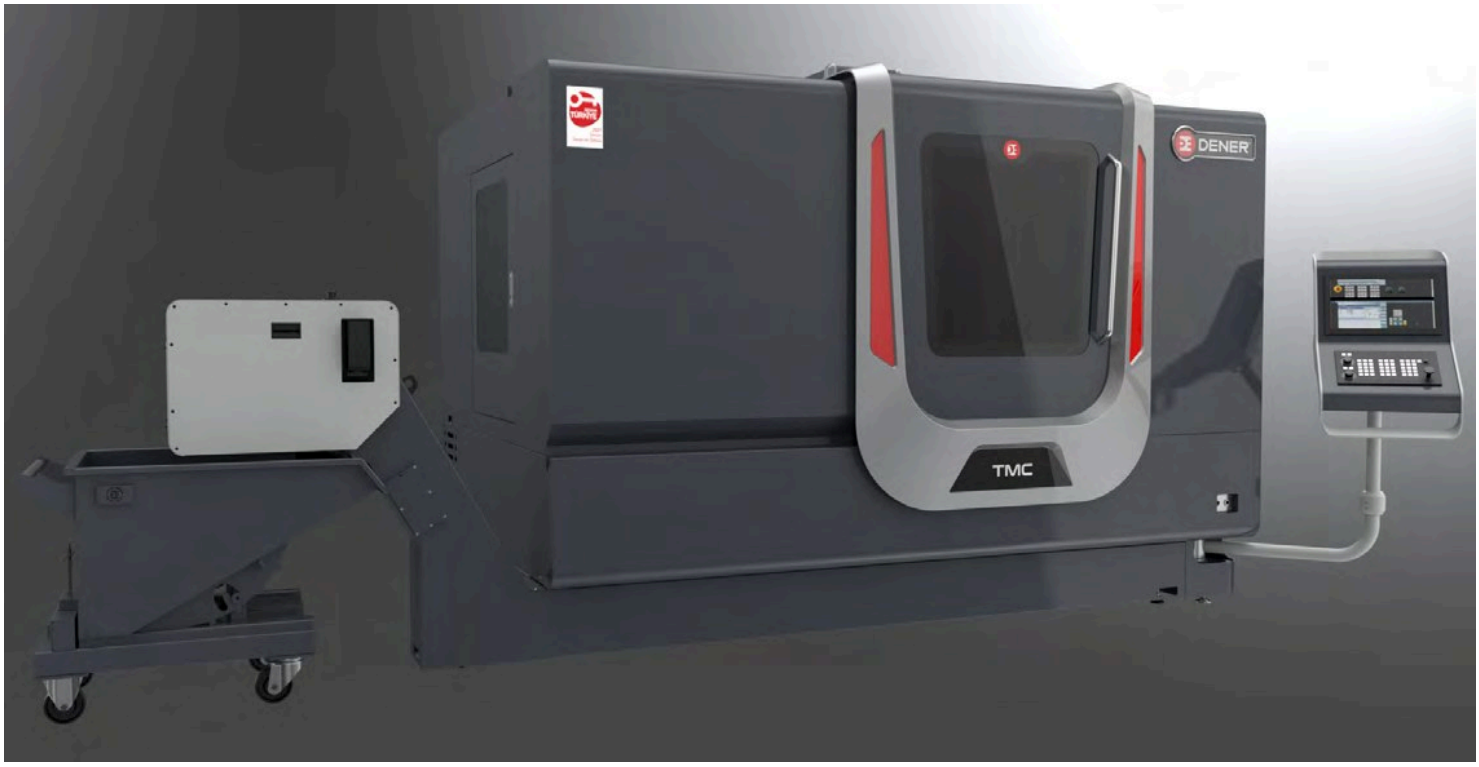
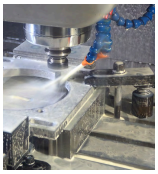
"THE FUTURE IS HERE" · SOLD & SERVICED IN CANADA BY NOVOTECH

MACHINING CENTRES
TMC · TDC SERIES

VERTICAL AND DOUBLE-COLUMN CNC MACHINING CENTRES









Since 1974
MANUFACTURING EXPERIENCE

33.5 – 161"
X-AXIS TRAVEL

BBT40 · BBT50
SPINDLE TAPER

Canada
PARTS & TECHNICAL SUPPORT

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MANUFACTURER · ESTABLISHED 1974

ABOUT DENER

Dener Makina was established in 1974 in the Kayseri Old Industrial Zone. Since 2000, the company has focused on the production of CNC Sheet Metal Processing Machines, manufacturing a wide range of products including Guillotine Shears, Hydraulic Press Brakes, Ball Screw Press Brakes, Servo-Electric Press Brakes, Plasma Cutting Machines, and Fiber Laser Cutting Machines.

In 2016, the company made a major investment by expanding its facilities to include 1,670,000 sq ft (155,000 m^2) of factory buildings and 290,000 sq ft (27,000 m^2) of social facilities on a 14,000,000 sq ft (1,300,000 m^2) site within the İncesu Organized Industrial Zone (O.S.B). In this new phase, Dener Makina began the production of CNC Metal Processing Machines such as Taksan Integrated Systems, Double Column Machining Centers, 5-Axis Simultaneous Machining Centers, Vertical Machining Centers, Lathes, and Grinding Machines.

With its extensive production range, over half a century of experience, skilled workforce, and multinational R&D center, Dener Makina holds a leading position globally by providing high-value industrial investment products in the fields of CNC Sheet Metal Processing and CNC Metal Processing Machines.

INCLUDED WITH EVERY NOVOTECH MACHINE

- ✓ Made in Turkey with European components
- ✓ CSA-approved electrical panel
- ✓ 1-year warranty
- ✓ **7-day money-back guarantee** (in-stock machines only)
- ✓ Spare parts & technical support readily available in Canada
- ✓ **Order Today with Just 10% Down — 0% Down Financing Available**
- ✓ **Installation and training available**

SOLD & SERVICED IN CANADA

by NovoTech Machine Tools — Mississauga, ON

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DENER MAKINA · İNCESU ORGANIZED INDUSTRIAL ZONE, KAYSERI

DENER PRODUCTION CAMPUS



Dener production campus, Kayseri — 1,670,000 sq ft (155,000 m^2) of factory buildings on a 14,000,000 sq ft (1,300,000 m^2) site

- SHEET PROCESSING MACHINES PRODUCTION PLANT
- MACHINE TOOLS PRODUCTION PLANT
- PRODUCTION PLANT
- SHEET WELDING FACTORY
- SHEET CLADDING FACTORY
- TECHNOLOGY CENTER
- ADMINISTRATIVE BUILDING
- SOCIAL FACILITIES
- MAIN ENTRANCE
- VISITOR PARKING AREA
- HELICOPTER PAD

52+
YEARS OF EXPERIENCE

100+
EXPORT COUNTRIES

14,000,000 sq ft
PRODUCTION CAMPUS (1,300,000 M^2)

TMC SERIES CNC VERTICAL MACHINING CENTRES

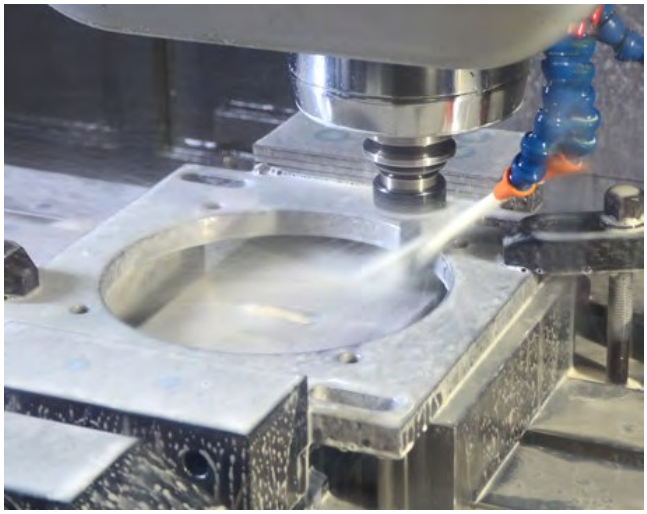
OVERVIEW



TMC CNC vertical machining centre

With CNC vertical machining centres, precision milling, drilling, pocketing and synchronized tapping without an adapter are possible, as well as the machining of forms defined by free-hand drawing.

- Specially manufactured precision, backlash-free double-nut ball screws
- Fast, serial automatic tool changer
- Precise, rigid direct-drive or belt-and-pulley driven spindles
- High-precision, economical linear motion
- Stable travel on linear or friction guideways
- Above-standard positioning and repeatability accuracy
- Stress-relieved Meehanite cast body
- Quiet, vibration-free, high cutting capability
- Fast service and spare parts support



TMC SERIES CNC VERTICAL MACHINING CENTRES

STANDARD / OPTIONAL EQUIPMENT

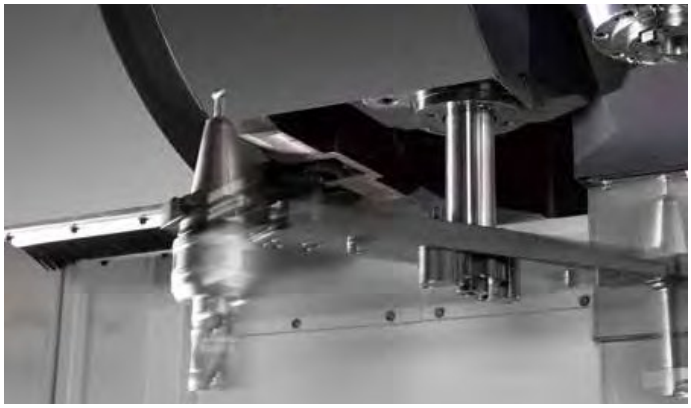
STANDARD EQUIPMENT

- Mitsubishi M80A-8A (15" touchscreen panel)
- Dialog programming (Navi-Mill)
- BT 40 / 10,000 rpm direct-drive spindle
- BT 40 arm-type 24-tool ATC
- C3 class double-nut ball screws
- Ball or roller linear guides
- Fully enclosed cabin
- Steel belt chip conveyor (with its bucket)
- Automatic lubrication system
- Spindle air blow system
- Coolant through nozzles around the spindle
- Oil and water separator unit
- Rigid tapping
- In-cabin chip washing system
- Electrical cabinet cooling system
- 3-phase 380 / 200 V – 50 / 60 Hz external transformer
- In-cabin lighting and alarm / warning lamps
- Front door safety switch
- Air and water gun
- Coolant system
- Portable handwheel
- 4th axis preparation
- Automatic power on / off
- Levelling bolts and plates
- Tool bag & tools
- CE

OPTIONAL EQUIPMENT

- FANUC 0i MF (15" LCD colour screen panel)
- BT 40 / 12,000 rpm and BT 40 / 15,000 rpm direct-drive spindles
- Through-tool coolant (435 psi, 30 bar)
- Air conditioner for the electrical cabinet
- CNC rotary table with the 4th axis
- Tool measuring system
- Workpiece measuring system
- Automatic door





TMC SERIES CNC VERTICAL MACHINING CENTRES

BODY STRUCTURE AND WORKING AREA

HIGH SPEED

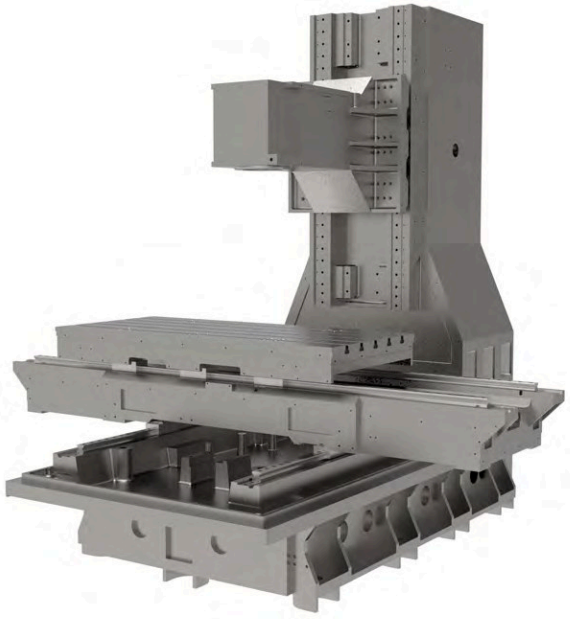
HIGH DURABILITY

PRECISION

RIGID AND OPTIMIZED MASSIVE CAST BODY

LONG LIFE

VIBRATION DAMPING



The cast column, base and table

WORKING AREA

Stroke	TMC 850	TMC 1000	TMC 1200	TMC 1300	TMC 1600
Stroke X / Y / Z	33.5 / 20.1 / 22″ (850 / 510 / 560 mm)	40.2 / 20.1 / 22″ (1,020 / 510 / 560 mm)	47.2 / 23.6 / 23.6″ (1,200 / 600 / 600 mm)	51.2 / 27.6 / 27.6″ (1,300 / 700 / 700 mm)	63 / 27.6 / 27.6″ (1,600 / 700 / 700 mm)

IDLE AXIS SPEEDS

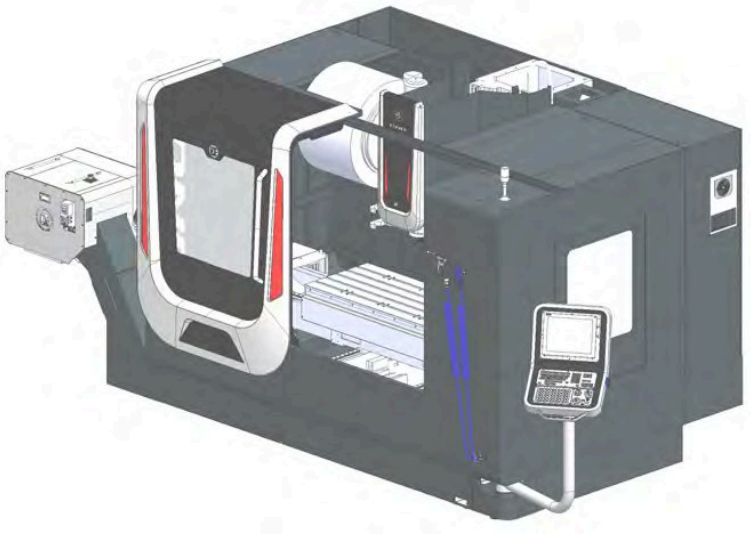
Rapid	TMC 850	TMC 1000	TMC 1200	TMC 1300	TMC 1600
Rapid X / Y / Z	118 / 118 / 118 ft/min (36 / 36 / 36 m/min)	118 / 118 / 118 ft/min (36 / 36 / 36 m/min)	118 / 118 / 118 ft/min (36 / 36 / 36 m/min)	118 / 118 / 118 ft/min (36 / 36 / 36 m/min)	118 / 118 / 118 ft/min (36 / 36 / 36 m/min)

TMC SERIES CNC VERTICAL MACHINING CENTRES

PART LOADING AND TABLE DIMENSIONS

PART LOADING

Capacity	TMC 850	TMC 1000	TMC 1200	TMC 1300	TMC 1600
Load capacity	1,102 lb (500 kg)	1,102 lb (500 kg)	2,205 lb (1,000 kg)	2,646 lb (1,200 kg)	2,646 lb (1,200 kg)



Loading a part by crane

TABLE DIMENSIONS

Table	TMC 850	TMC 1000	TMC 1200	TMC 1300	TMC 1600
Table L × W	39.4 × 19.7″ (1,000 × 500 mm)	47.2 × 19.7″ (1,200 × 500 mm)	51.2 × 23.6″ (1,300 × 600 mm)	57.1 × 27.6″ (1,450 × 700 mm)	68.9 × 27.6″ (1,750 × 700 mm)

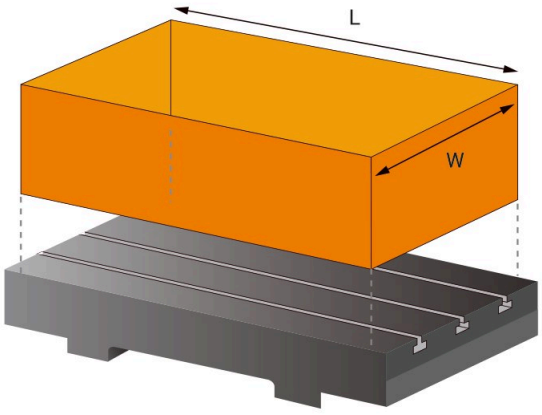
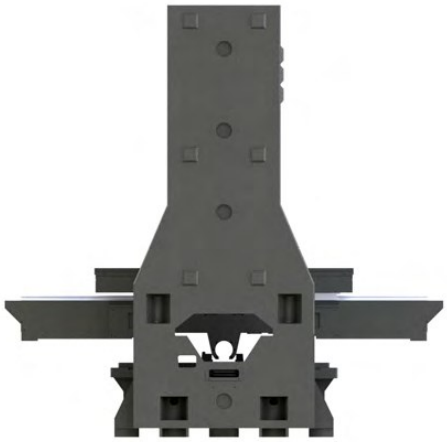
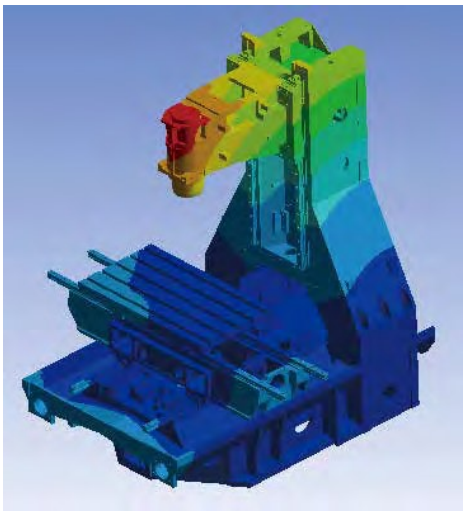


Table length L and width W

TMC SERIES CNC VERTICAL MACHINING CENTRES

STRUCTURE



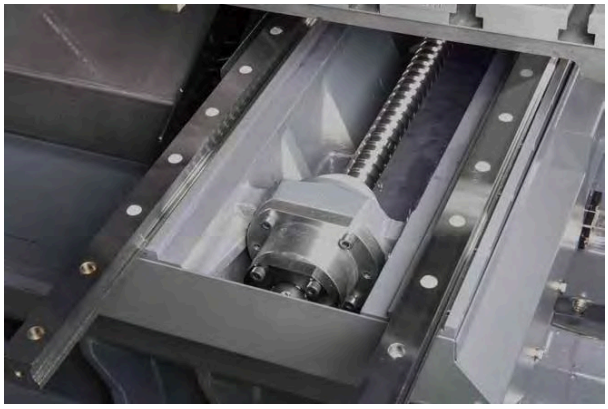


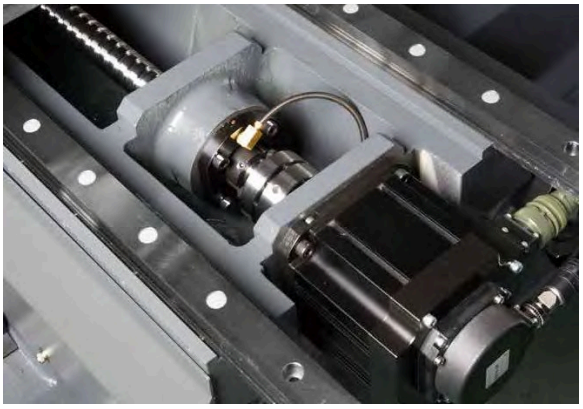
RIGID STRUCTURE

Excellent vibration damping and cutting-load distribution thanks to the combined design of the deltoid column and the main body.

FINITE ELEMENT METHOD

The optimum body structure obtained using the finite element method.





ULTRA-PRECISE LINEAR EQUIPMENT

Ultra-precise, ground, preloaded and backlash-free ball screws and linear bearing elements for precise positioning, repeatability, minimum friction, rigid support and quiet operation.

DIRECT-DRIVE SERVO MOTORS

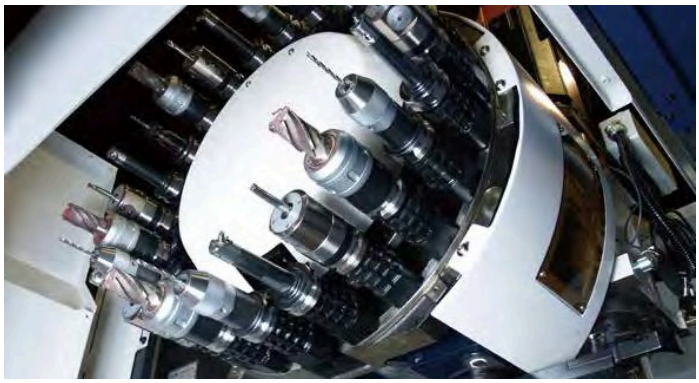
Direct connection between the motor and the ball screw for backlash-free, precise motion.

TMC SERIES CNC VERTICAL MACHINING CENTRES

TOOL CHANGE AND SPINDLE

TOOL CHANGE SYSTEM

Fast, stable and functional tool change with the 24-tool, cam-type, inverter-controlled ATC.





DIRECT-DRIVE SPINDLE

Powerful machining capability with high heat dissipation, low vibration, quiet operation and easy maintenance.



SPINDLE COOLING SYSTEM

Automatically controlled cooling system to prevent thermal deformation and extend the spindle's life.





THROUGH-TOOL COOLANT (OPTION)

Excellent cutting capability and long tool life with coolant through the tool.



TMC SERIES CNC VERTICAL MACHINING CENTRES

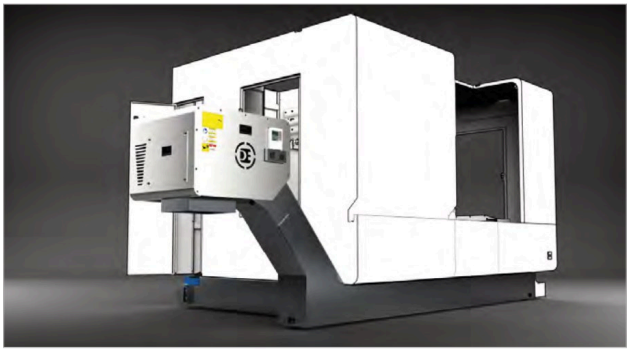
CHIP CONVEYOR, COOLANT AND CONTROL UNIT

CHIP CONVEYOR

Trouble-free transport and discharge of the long, coarse and spiral chips of metal, fiber, teflon, delrin and similar materials.

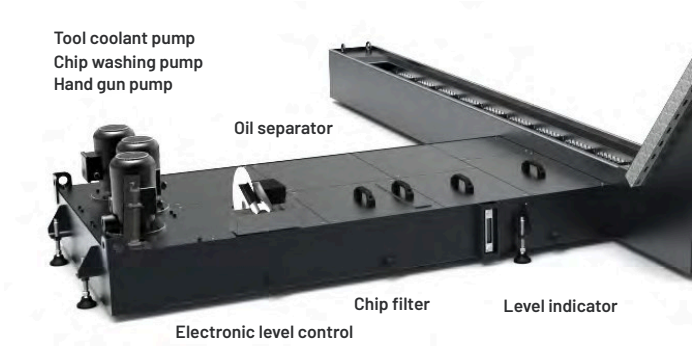






COOLANT TANK

Optimum cooling and easy maintenance with a 132 US gal (500 L), self-filtering, functional tank design.



CHIP CART

Easy storage and discharge of all kinds of chips, with its tilting and braking features.



CONTROL UNIT

Easy to use, with up-to-date technological modules, an operator-friendly interface and an ergonomic design.





TMC SERIES CNC VERTICAL MACHINING CENTRES

TECHNICAL DATA · 1 OF 4

Model	TMC 850	TMC 1000	TMC 1200	TMC 1300	TMC 1600
MOVEMENTS					
X-axis travel	33.5″ (850 mm)	40.2″ (1,020 mm)	47.2″ (1,200 mm)	51.2″ (1,300 mm)	63″ (1,600 mm)
Y-axis travel	20.1″ (510 mm)	20.1″ (510 mm)	23.6″ (600 mm)	27.6″ (700 mm)	27.6″ (700 mm)
Z-axis travel	22″ (560 mm)	22″ (560 mm)	23.6″ (600 mm)	27.6″ (700 mm)	27.6″ (700 mm)
Ball screws X / Y	C3	C3	C3	C3	C3
Distance from the column face to the spindle centre	5.9 / 28″ (150 / 710 mm)	5.9 / 28″ (150 / 710 mm)	5.9 / 29.5″ (150 / 750 mm)	5.1 / 32.7″ (130 / 830 mm)	5.1 / 32.7″ (130 / 830 mm)
Guideway type	Roller type	Roller type	Roller type	Roller type	Roller type
Guideway size X / Y / Z	1.18 / 1.38 / 1.38″ (30 / 35 / 35 mm)	1.18 / 1.38 / 1.38″ (30 / 35 / 35 mm)	1.38 / 1.77 / 1.77″ (35 / 45 / 45 mm)	1.38 / 1.77 / 1.77″ (35 / 45 / 45 mm)	1.38 / 1.77 / 1.77″ (35 / 45 / 45 mm)
WORK TABLE					
Table size (X direction)	39.4″ (1,000 mm)	47.2″ (1,200 mm)	51.2″ (1,300 mm)	57.1″ (1,450 mm)	68.9″ (1,750 mm)
Table size (Y direction)	19.7″ (500 mm)	19.7″ (500 mm)	23.6″ (600 mm)	27.6″ (700 mm)	27.6″ (700 mm)
T-slot size	0.71″ (18 mm)	0.71″ (18 mm)	0.71″ (18 mm)	0.71″ (18 mm)	0.71″ (18 mm)
Number of T-slots	5	5	5	5	5
T-slot spacing	3.9″ (100 mm)	3.9″ (100 mm)	3.9″ (100 mm)	4.9″ (125 mm)	4.9″ (125 mm)
Table load capacity	1,102 lb (500 kg)	1,102 lb (500 kg)	2,205 lb (1,000 kg)	2,646 lb (1,200 kg)	2,646 lb (1,200 kg)
SPINDLE					
Spindle taper (7/24 taper)	BBT40	BBT40	BBT40	BBT40	BBT40
Spindle drive	Direct	Direct	Direct	Direct	Direct
FEEDS					
Rapid traverse X / Y	118 ft/min (36 m/min)	118 ft/min (36 m/min)	118 ft/min (36 m/min)	118 ft/min (36 m/min)	118 ft/min (36 m/min)
Rapid traverse Z	118 ft/min (36 m/min)	118 ft/min (36 m/min)	118 ft/min (36 m/min)	118 ft/min (36 m/min)	118 ft/min (36 m/min)
Cutting feed	3.3 – 49.2 ft/min (1 – 15 m/min)	3.3 – 49.2 ft/min (1 – 15 m/min)	3.3 – 49.2 ft/min (1 – 15 m/min)	3.3 – 49.2 ft/min (1 – 15 m/min)	3.3 – 49.2 ft/min (1 – 15 m/min)
Tool magazine capacity	24	24	24	24	24

TMC SERIES CNC VERTICAL MACHINING CENTRES

TECHNICAL DATA · 2 OF 4

Model	TMC 850	TMC 1000	TMC 1200	TMC 1300	TMC 1600
TOOL MAGAZINE					
Tool change method	Arm type	Arm type	Arm type	Arm type	Arm type
Tool shank type	BBT	BBT	BBT	BBT	BBT
Tool change time (tool to tool)	1.3 s	1.3 s	1.3 s	1.3 s	1.3 s
Tool change time (chip to chip)	3.1 s	3.1 s	3.7 s	3.7 s	3.7 s
Pull stud angle	45°	45°	45°	45°	45°
Max. tool diameter	Ø3.9 / Ø5.9″ (Ø100 / Ø150 mm)	Ø3.9 / Ø5.9″ (Ø100 / Ø150 mm)	Ø3.9 / Ø5.9″ (Ø100 / Ø150 mm)	Ø3.9 / Ø5.9″ (Ø100 / Ø150 mm)	Ø3.9 / Ø5.9″ (Ø100 / Ø150 mm)
Max. tool length	11.8″ (300 mm)	11.8″ (300 mm)	11.8″ (300 mm)	11.8″ (300 mm)	11.8″ (300 mm)
Max. tool weight	15 lb (7 kg)	15 lb (7 kg)	15 lb (7 kg)	15 lb (7 kg)	15 lb (7 kg)
ACCURACY					
Positioning accuracy	±0.00016″ / 11.8″ (±0.004 / 300 mm)	±0.00016″ / 11.8″ (±0.004 / 300 mm)	±0.00016″ / 11.8″ (±0.004 / 300 mm)	±0.00016″ / 11.8″ (±0.004 / 300 mm)	±0.0002″ / 11.8″ (±0.005 / 300 mm)
Repeatability	±0.00008″ / 11.8″ (±0.002 / 300 mm)	±0.00008″ / 11.8″ (±0.002 / 300 mm)	±0.00008″ / 11.8″ (±0.002 / 300 mm)	±0.00008″ / 11.8″ (±0.002 / 300 mm)	±0.00008″ / 11.8″ (±0.002 / 300 mm)
COOLANT — CAPACITY, MOTOR, PRESSURE					
Coolant pump	0.74 HP (44 psi) (0.55 kW (3 bar))	0.74 HP (44 psi) (0.55 kW (3 bar))	0.74 HP (44 psi) (0.55 kW (3 bar))	0.74 HP (44 psi) (0.55 kW (3 bar))	0.74 HP (44 psi) (0.55 kW (3 bar))
Coolant pump for the gun	0.5 HP (29 psi) (0.37 kW (2 bar))	0.5 HP (29 psi) (0.37 kW (2 bar))	0.5 HP (29 psi) (0.37 kW (2 bar))	0.5 HP (29 psi) (0.37 kW (2 bar))	0.5 HP (29 psi) (0.37 kW (2 bar))
Coolant pump for chip cleaning	1.48 HP (36 psi) (1.1 kW (2.5 bar))	1.48 HP (36 psi) (1.1 kW (2.5 bar))	1.48 HP (36 psi) (1.1 kW (2.5 bar))	1.48 HP (36 psi) (1.1 kW (2.5 bar))	1.48 HP (36 psi) (1.1 kW (2.5 bar))
Coolant tank capacity	92 US gal (350 L)	92 US gal (350 L)	106 US gal (400 L)	106 US gal (400 L)	106 US gal (400 L)
MACHINE DIMENSIONS (MACHINE WORKING AREA)					
Length	156.1″ (3,965 mm)	169.3″ (4,300 mm)	184.4″ (4,685 mm)	194.7″ (4,945 mm)	224″ (5,690 mm)
Width	86.6″ (2,200 mm)	86.6″ (2,200 mm)	94.5″ (2,400 mm)	93.3″ (2,370 mm)	94.5″ (2,400 mm)
Height	125.4″ (3,185 mm)	125.4″ (3,185 mm)	123.8″ (3,145 mm)	129.1″ (3,280 mm)	129.1″ (3,280 mm)
Weight (approx.)	14,330 lb (6,500 kg)	16,535 lb (7,500 kg)	17,637 lb (8,000 kg)	20,944 lb (9,500 kg)	22,046 lb (10,000 kg)

The values in the table may change with renewals and revisions of the machines.

TMC SERIES CNC VERTICAL MACHINING CENTRES

TECHNICAL DATA · 3 OF 4

MITSUBISHI CONTROL UNIT (STANDARD)

Model	TMC 850 (M)	TMC 1000 (M)	TMC 1200 (M)	TMC 1300 (M)	TMC 1600 (M)
Screen	10.4″ / 15″	10.4″ / 15″	15″	15″	15″
SPINDLE MOTOR					
Code	SJ-DG11/100-03T	SJ-DG11/100-03T	SJ-DG11/120-03T	SJ-DG15/120-02T-K-S	SJ-DG15/120-02T-K-S
Speed	10,000 rpm	10,000 rpm	12,000 rpm	12,000 rpm	12,000 rpm
CTS readiness	–*	–*	+	+	+
Torque (continuous / max.)	35.2 / 84.8 lbf·ft (47.7 / 115 N·m)	35.2 / 84.8 lbf·ft (47.7 / 115 N·m)	35.2 / 84.8 lbf·ft (47.7 / 115 N·m)	57.4 / 140.9 lbf·ft (77.8 / 191 N·m)	57.4 / 140.9 lbf·ft (77.8 / 191 N·m)
Power (continuous / max.)	10.1 / 24.1 HP (7.5 / 18 kW)	10.1 / 24.1 HP (7.5 / 18 kW)	10.1 / 24.1 HP (7.5 / 18 kW)	14.8 / 40.2 HP (11 / 30 kW)	14.8 / 40.2 HP (11 / 30 kW)
AXIS MOTORS					
Code X / Y	HG224S-D48	HG224S-D48	HG354S-D48	HG354S-D48	HG354S-D48
Torque X / Y (continuous / max.)	5.2 / 34.3 lbf·ft (7 / 46.5 N·m)	5.2 / 34.3 lbf·ft (7 / 46.5 N·m)	8.2 / 55.3 lbf·ft (11.1 / 75 N·m)	8.2 / 55.3 lbf·ft (11.1 / 75 N·m)	10.5 / 90 lbf·ft (14.3 / 122 N·m)
Power X / Y	3 HP (2.2 kW)	3 HP (2.2 kW)	4.7 HP (3.5 kW)	4.7 HP (3.5 kW)	4.7 HP (3.5 kW)
Code Z	HG354BS-D48	HG354BS-D48	HG354BS-D48	HG453BS-D48	HG453BS-D48
Torque Z (continuous / max.)	8.2 / 55.3 lbf·ft (11.1 / 75 N·m)	8.2 / 55.3 lbf·ft (11.1 / 75 N·m)	8.2 / 55.3 lbf·ft (11.1 / 75 N·m)	10.5 / 90 lbf·ft (14.3 / 122 N·m)	10.5 / 90 lbf·ft (14.3 / 122 N·m)
Power Z	4.7 HP (3.5 kW)	4.7 HP (3.5 kW)	4.7 HP (3.5 kW)	6 HP (4.5 kW)	6 HP (4.5 kW)
AIR AND POWER CONNECTION					
Air pressure	73 / 87 psi (5.0 / 6.0 bar)	73 / 87 psi (5.0 / 6.0 bar)	73 / 87 psi (5.0 / 6.0 bar)	73 / 87 psi (5.0 / 6.0 bar)	73 / 87 psi (5.0 / 6.0 bar)
Power connection	32 kVA	32 kVA	32 kVA	43 kVA	43 kVA

** CTS = SJ-DG11/120-03T-S (12,000 rpm).*

TMC SERIES CNC VERTICAL MACHINING CENTRES

TECHNICAL DATA · 4 OF 4

FANUC CONTROL UNIT (OPTION)

Model	TMC 850 (F)	TMC 1000 (F)	TMC 1200 (F)	TMC 1300 (F)	TMC 1600 (F)
Screen	10.4″	10.4″	15″	15″	15″
SPINDLE MOTOR					
Code	βiI 8/12000-B	βiI 8/12000-B	βiIT 12/10000-B	βiIT 12/10000-B	βiIT 12/10000-B
Speed	12,000 rpm	12,000 rpm	10,000 rpm	10,000 rpm	10,000 rpm
CTS readiness	-**	-**	+	+	+
Torque (continuous / max.)	26.4 / 77.4 lbf·ft (35.8 / 105 N·m)	26.4 / 77.4 lbf·ft (35.8 / 105 N·m)	38.7 / 95.9 lbf·ft (52.5 / 130 N·m)	38.7 / 95.9 lbf·ft (52.5 / 130 N·m)	38.7 / 95.9 lbf·ft (52.5 / 130 N·m)
Power (continuous / max.)	10.1 / 22.1 HP (7.5 / 16.5 kW)	10.1 / 22.1 HP (7.5 / 16.5 kW)	14.8 / 27.4 HP (11 / 20.4 kW)	14.8 / 27.4 HP (11 / 20.4 kW)	14.8 / 27.4 HP (11 / 20.4 kW)
AXIS MOTORS					
Code X / Y	βiS 22/3000-B	βiS 22/3000-B	βiS 22/3000-B	βiS 22/3000-B	βiS 22/3000-B
Torque X / Y (continuous / max.)	14.8 / 33.2 lbf·ft (20 / 45 N·m)	14.8 / 33.2 lbf·ft (20 / 45 N·m)	14.8 / 33.2 lbf·ft (20 / 45 N·m)	14.8 / 33.2 lbf·ft (20 / 45 N·m)	14.8 / 33.2 lbf·ft (20 / 45 N·m)
Power X / Y	4 HP (3 kW)	4 HP (3 kW)	4 HP (3 kW)	4 HP (3 kW)	4 HP (3 kW)
Code Z	βiS 22/3000-B	βiS 22/3000-B	βiS 22/3000-B	βiS 40/2000-B	βiS 40/2000-B
Torque Z (continuous / max.)	14.8 / 33.2 lbf·ft (20 / 45 N·m)	14.8 / 33.2 lbf·ft (20 / 45 N·m)	14.8 / 33.2 lbf·ft (20 / 45 N·m)	26.6 / 66.4 lbf·ft (36 / 90 N·m)	26.6 / 66.4 lbf·ft (36 / 90 N·m)
Power Z	4 HP (3 kW)	4 HP (3 kW)	4 HP (3 kW)	4 HP (3 kW)	4 HP (3 kW)
AIR AND POWER CONNECTION					
Air pressure	73 / 87 psi (5.0 / 6.0 bar)	73 / 87 psi (5.0 / 6.0 bar)	73 / 87 psi (5.0 / 6.0 bar)	73 / 87 psi (5.0 / 6.0 bar)	73 / 87 psi (5.0 / 6.0 bar)
Power connection	32 kVA	32 kVA	32 kVA	32 kVA	32 kVA

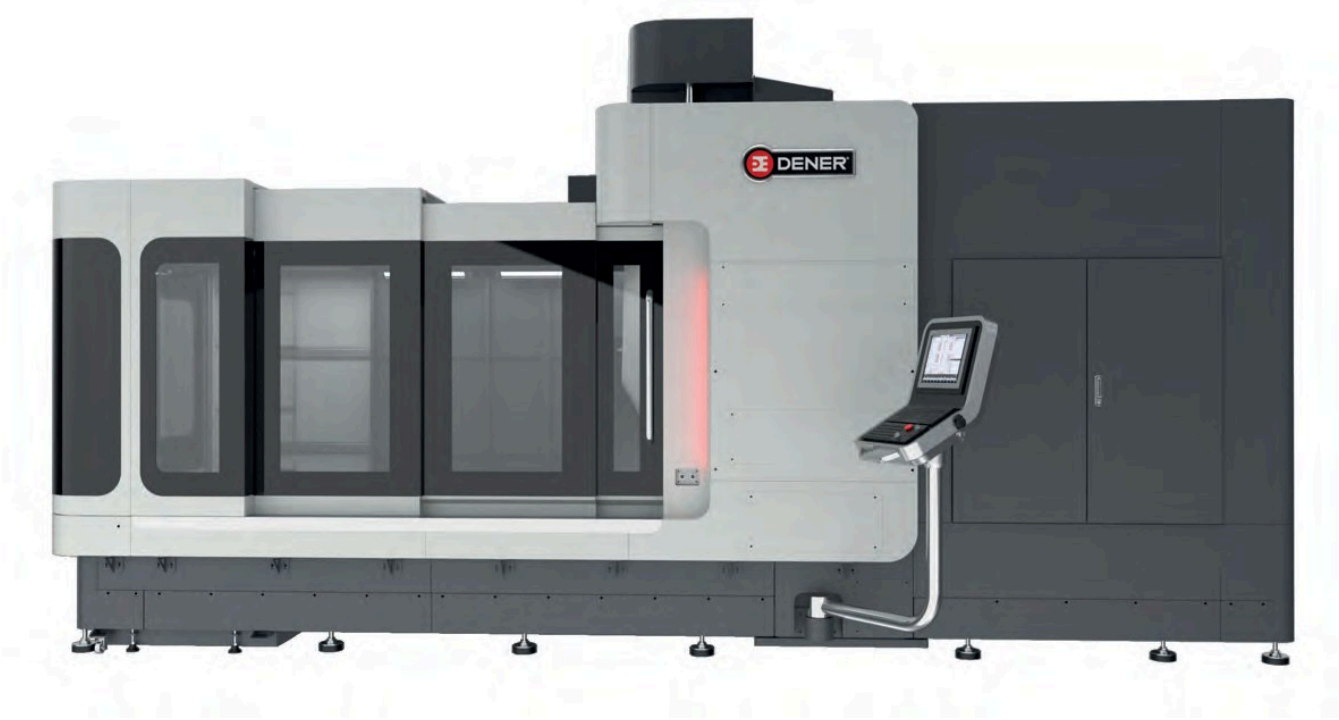
*** CTS = βiIT 12/10000-B (10,000 rpm). The values in the table may change with renewals and revisions of the machines. Custom voltage available: 208 V, 240 V, 480 V or 600 V 3-phase. CSA-approved electrical panel.*

INCLUDED WITH EVERY NOVOTECH MACHINE

- Made in Turkey with European components
- CSA-approved electrical panel
- 1-year warranty
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DENER · CNC MACHINING CENTRE

DOUBLE-COLUMN MACHINING CENTRE



TDC double-column machining centre

2×
MOTOR POWER COMPARED TO COMPETITORS

Long
DISTANCE BETWEEN THE TWO COLUMNS

The Dener double-column CNC machining centre increases the competitiveness of your business by combining creative design, technological superiority and productivity. With this machine, which meets your precision machining needs at the highest level, you can produce quickly and efficiently without compromising on quality.

TDC SERIES DOUBLE-COLUMN CNC MACHINING CENTRES

CONTROL UNIT, CHIP CONVEYOR AND TOOL CHANGE



CONTROL UNIT

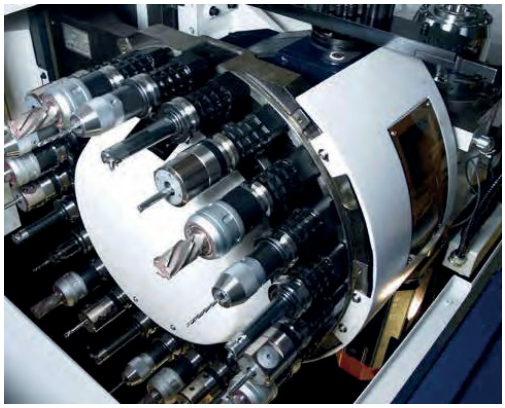
The Mitsubishi control unit makes your production and machining processes more efficient and flexible with its powerful performance, advanced features and reliability.

- Advanced control capabilities
- Fast response time
- Comprehensive connection options
- User-friendly interface



CHIP CONVEYOR

The chip conveyor controls your machining processes and keeps them running steadily, increasing your machining efficiency. Thanks to this solution, the preparation time for complex operations is shorter, and your ability to handle a wide variety of workpieces grows.

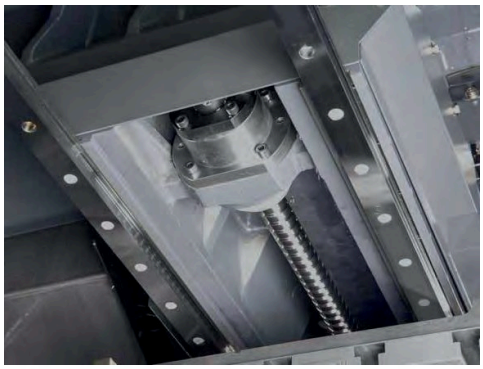


TOOL CHANGE

The tool change system optimizes efficiency with fast, automatic tool change. Fast, automatic tool change shortens machining times and raises productivity while keeping the quality.

TDC SERIES DOUBLE-COLUMN CNC MACHINING CENTRES

LINEAR EQUIPMENT, COOLANT AND SPINDLE



PRECISE LINEAR EQUIPMENT

Precise linear equipment offers excellent accuracy, smooth movement and long life. It keeps the precision of your machining processes stable and so gives higher-quality workpieces.



COOLANT TANK

The coolant tank optimizes your machining processes while extending tool life, keeping the machining conditions efficient and contributing to excellent results.



DIRECT-DRIVE SPINDLE

High torque, fast acceleration and excellent precision: this advanced spindle technology takes the performance of your CNC machining centre to new levels, bringing speed and precision together for outstanding machining results.

RPM	BBT-50	CTS
6,000 rpm	○	○
8,000 rpm	○	○
10,000 rpm	●	●

std. ● · opt. ○



SPINDLE COOLING SYSTEM

Extends tool life by controlling the temperature, and keeps the machining process stable. This cooling solution maximizes the performance of your CNC machining centre.



THROUGH-TOOL COOLANT

Automatically controlled cooling system to prevent thermal deformation and extend the spindle's life.

TDC SERIES DOUBLE-COLUMN CNC MACHINING CENTRES

TECHNICAL DATA

TDC double column	TDC 2112	TDC 3116	TDC 4120
X axis	82.7" (2,100 mm)	122" (3,100 mm)	161.4" (4,100 mm)
Y axis	48" (1,220 mm)	63" (1,600 mm)	79.5" (2,020 mm)
Z axis	23.6" (opt. 31.5") (600 mm, opt. 800 mm)	40.2" (1,020 mm)	40.2" (1,020 mm)
Distance between the two columns	55.1" (1,400 mm)	66.9" (1,700 mm)	82.7" (2,100 mm)
Table size	78.7 × 47.2" (2,000 × 1,200 mm)	118.1 × 59.1" (3,000 × 1,500 mm)	157.5 × 70.9" (4,000 × 1,800 mm)
T-slots (number × slot width × distance)	7 × 0.87" × 5.9" (7 × 22 × 150 mm)	9 × 0.87" × 5.9" (9 × 22 × 150 mm)	9 × 0.87" × 5.9" (9 × 22 × 150 mm)
Max. table load	8,818 lb (4,000 kg)	22,046 lb (10,000 kg)	26,455 lb (12,000 kg)
Spindle taper	BBT50	BBT50	BBT50
Spindle speed	10,000 rpm	10,000 rpm	10,000 rpm
Spindle motor	29.5 / 34.9 HP (22 / 26 kW)	29.5 / 34.9 HP (22 / 26 kW)	29.5 / 34.9 HP (22 / 26 kW)
Rapid traverse X / Y / Z	39 / 39 / 39 ft/min (12 / 12 / 12 m/min)	39 / 39 / 39 ft/min (12 / 12 / 12 m/min)	39 / 39 / 39 ft/min (12 / 12 / 12 m/min)
Cutting feed	3.3 – 32.8 ft/min (1 – 10 m/min)	3.3 – 32.8 ft/min (1 – 10 m/min)	3.3 – 32.8 ft/min (1 – 10 m/min)
Axis motors X, Y, Z	12.1 / 6 / 6 HP (9 / 4.5 / 4.5 kW)	12.1 / 9.4 / 9.4 HP (9 / 7 / 7 kW)	12.1 / 9.4 / 9.4 HP (9 / 7 / 7 kW)
Tool capacity	Arm type, 24 tools	Chain type, 40 tools	Chain type, 40 tools
Coolant tank capacity	79 US gal (300 L)	211 US gal (800 L)	211 US gal (800 L)
Guideway types	X, Y, Z roller linear guides	X, Y, Z roller linear guides	X, Y, Z roller linear guides
Control unit	Mitsubishi (Fanuc option)	Mitsubishi (Fanuc option)	Mitsubishi (Fanuc option)
Positioning accuracy X / Y / Z	±0.0003" (±0.008 mm)	±0.0003" (±0.008 mm)	±0.0003" (±0.008 mm)
Repeatability X / Y / Z	±0.00016" (±0.004 mm)	±0.00016" (±0.004 mm)	±0.00016" (±0.004 mm)
Guarding	Semi-open enclosure	Semi-open enclosure	Semi-open enclosure
Width	143.3" (3,640 mm)	170.9" (4,340 mm)	186.6" (4,740 mm)
Length	234.8" (5,965 mm)	327.4" (8,315 mm)	406.1" (10,315 mm)
Height	132.8" (3,372 mm)	172.8" (4,390 mm)	172.8" (4,390 mm)
Weight (approx.)	46,297 lb (21,000 kg)	70,548 lb (32,000 kg)	88,185 lb (40,000 kg)

Custom voltage available: 208 V, 240 V, 480 V or 600 V 3-phase. CSA-approved electrical panel.

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Hydraulic press brake



Tandem press brake



Servo-electric press brake



Ball-screw press brake

PANEL BENDER



Panel bender

SHEARS



CNC hydraulic shear



NC hydraulic shear

LASER CUTTING MACHINES



GFL fiber laser cutting machine



Smart fiber laser cutting machine



Tube profile laser cutting machine

MACHINE TOOLS



CNC lathe



CNC vertical milling machine



Double-column milling machine



5-axis machining center



NC grinding machine

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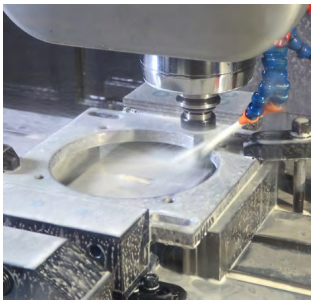
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Specifications subject to change without notice. Imperial values are converted from the manufacturer's metric data and rounded: plate thicknesses in common fractions (1/2, 1/4, 1/8, 1/16). Model numbers are manufacturer designations. Photos may show optional equipment.